

Quick Sort

27 March 2023 16:11

pivot

66

left

right

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
66	33	99	11	77	22	55	44	88

i

j

66 <= pivot (66) ? Yes i++

33 <= 66 ? Yes i++

99 <= 66 ? No --> swap with other element from the right.

88 > 66 ? Yes j--

44 > 66 ? No swap 99 with 44

pivot

66

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
66	33	44	11	77	22	55	99	88

i

j

44 <= 66 ? Yes i++

11 <= 66 ? Yes i++

77 <= 66 ? No --> swap with other element from right

99 > 66 ? Yes j--

55 > 66 ? No swap with i

Pivot

66

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
66	33	44	11	55	22	77	99	88

j

55 <= 66 ? Yes i++

22 <= 66 ? Yes i++

77 <= 66 ? No --> swap

77 > 66 ? Yes j--

i and j cross --> swap

j with left element

left

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
22	33	44	11	55	66	77	99	88

Left partitioning --> left to j-1

Right partitioning --> j+1 to right

pivot

22

left

right

[0]	[1]	[2]	[3]	[4]
22	33	44	11	55

i

j

22 <= 22 ? Yes i++

33 <= 22 ? No --> swap

55 > 22 ? Yes --> j--

11 > 22 ? No --> swap with i

[0]	[1]	[2]	[3]	[4]
22	11	44	33	55

j

i

11 <= 22 ? Yes --> i++

44 <= 22 ? No --> swap

Pivot

77

left

right

[6]	[7]	[8]
77	99	88

j

i

77 <= 77 ? Yes i++

99 <= 77 ? No --> swap

88 > 77 ? Yes --> j--

99 > 77 ? Yes j--

i and j cross --> swap j with ele

[6]	[7]	[8]
77	99	88

Right partitioning

